CSEC Mathematics

May/June 2026 – Paper 2

Worked Solutions

Reviewed and published by Kerwin Springer at The Student Hub

SECTION I

Answer ALL questions.
All working must be clearly shown.

1. (a) (i) Write the following numbers correct to 1 significant figure.

0.367

$$0.367 \approx 0.4 \text{ to 1 sf}$$

3.94

$$3.94 \approx 4 \text{ to 1 sf}$$

(2 marks)

(ii) Hence, estimate the integer value of $\frac{\sqrt{1000} \times 0.367}{3.94}$.

$$\begin{aligned}\frac{\sqrt{1000} \times 0.367}{3.94} &\approx \frac{\sqrt{1000} \times 0.4}{4} \\ &= \sqrt{100} \\ &= 10\end{aligned}$$

(1 mark)

(b) Aiden, Brooke and Caden share berries in the ratio $3:2:5$. Aiden receives 12 more berries than Brooke. Calculate the TOTAL number of berries shared.

Difference:

$$\begin{aligned}\text{Aiden} - \text{Brooke} &= 3 \text{ parts} - 2 \text{ parts} \\ &= 1 \text{ part} \\ 1 \text{ part} &= 12\end{aligned}$$

$$\begin{aligned}\text{Total parts} &= 3 + 2 + 5 = 10 \text{ parts} \\ \text{Total berries} &= 10 \times 12 = 120 \text{ berries}\end{aligned}$$

(2 marks)

(c) In a sale, all prices are reduced by 15%.

(i) Blake buys a book which has an original price of \$16.40. Calculate how much he pays.

$$\begin{aligned}\text{Sale price} &= 85\% \text{ of original price} \\ &= 0.85 \times 16.40 \\ &= \$13.94\end{aligned}$$

(2 marks)

(ii) Jude pays \$27.20 for a toy. Calculate the original price of the toy.

$$\begin{aligned} 85\% &= 27.20 \\ 1\% &= \frac{27.20}{85} \\ \text{Original price } (100\%) &= \frac{27.20}{85} \times 100 \\ &= \$32.00 \end{aligned}$$

(2 marks)

Total 9 marks

2. (a) Given that $A = \dfrac{p^2 - q}{2r}$,

(i) find the value of A when $p = 2$, $q = -1$ and $r = 1$.

$$\begin{aligned} A &= \frac{p^2 - q}{2r} \\ &= \frac{(2)^2 - (-1)}{2 \times (1)} \\ &= \frac{4 + 1}{2} \\ &= 2.5 \end{aligned}$$

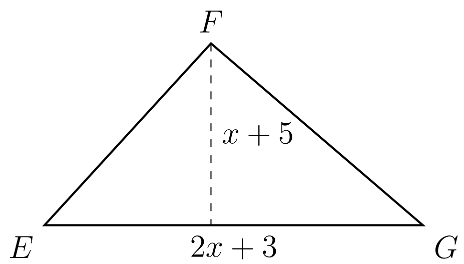
(1 mark)

(ii) Express q in terms of A, p and r.

$$\begin{aligned} A &= \frac{p^2 - q}{2r} \\ 2Ar &= p^2 - q \\ q &= p^2 - 2Ar \end{aligned}$$

(2 marks)

(b) In triangle EFG, $EG = (2x + 3)$ cm and the perpendicular height from F to EG is $(x + 5)$ cm. Given that the area of triangle EFG $= 50$ cm^2,



(i) Show that $2x^2 + 13x - 85 = 0$.

$$\begin{aligned} \text{Area} &= \tfrac{1}{2} \times \text{base} \times \text{height} \\ 50 &= \tfrac{1}{2}(2x + 3)(x + 5) \\ 100 &= (2x + 3)(x + 5) \\ 100 &= 2x^2 + 10x + 3x + 15 \\ 100 &= 2x^2 + 13x + 15 \\ 2x^2 + 13x - 85 &= 0 \end{aligned}$$

(3 marks)

(ii) Find the value of x, correct to 2 decimal places.

Using the Quadratic Formula with $a = 2$, $b = 13$, $c = -85$:

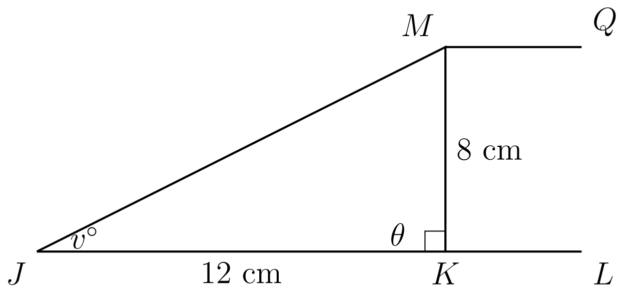
$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-13 \pm \sqrt{13^2 - 4(2)(-85)}}{2(2)} \\ &= \frac{-13 \pm \sqrt{849}}{4} \end{aligned}$$

$$x = 4.03 \quad \text{or} \quad x = -10.53$$

Since length must be positive, $x = 4.03$. *(3 marks)*

Total 9 marks

3. In the diagram, JKL is a straight line and JMK is a right-angled triangle. The line segment JM is parallel to KQ, $JK = 12$ cm and $MK = 8$ cm.



(a) Determine

(i) $\sin v°$.

$$\sin v° = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{MK}{JK}$$
$$= \frac{8}{12} = \frac{2}{3}$$

(1 mark)

(ii) The length of JM.

$$JM^2 + MK^2 = JK^2$$
$$JM^2 + (8)^2 = 12^2$$
$$JM^2 = 144 - 64 = 80$$
$$JM = \sqrt{80} = 8.94 \text{ cm to 3 sf}$$

(2 marks)

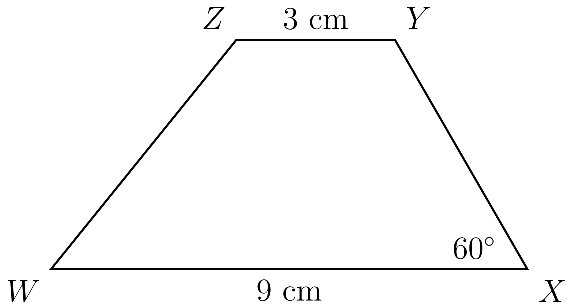
(iii) $\angle MKL$.

$$\cos\theta = \frac{\text{adj}}{\text{hyp}} = \frac{8}{12}$$
$$\theta = \cos^{-1}\left(\frac{8}{12}\right) = 48.2°$$

$$\angle MKL = 180° - 48.2° = 131.8°$$

(2 marks)

(b) Construct a trapezium WXYZ, with $WX = 9$ cm, $XY = 5$ cm, $YZ = 3$ cm, $\angle WXY = 60°$ and $WX \parallel YZ$.



(4 marks)

Total 9 marks

4. (a) The coordinates of A and B are $(-3, -4)$ and $(2, 6)$ respectively. For the line segment AB, determine:

(i) Midpoint, M.

$$\begin{aligned} M &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \\ &= \left(\frac{-3+2}{2}, \frac{-4+6}{2}\right) \\ &= \left(-\tfrac{1}{2}, 1\right) \end{aligned}$$

(2 marks)

(ii) Gradient, m.

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{6 - (-4)}{2 - (-3)} \\ &= \frac{10}{5} = 2 \end{aligned}$$

(2 marks)

(iii) Equation of its perpendicular bisector.

Gradient of $AB = 2$, so gradient of perpendicular bisector $= -\frac{1}{2}$.

Using midpoint $\left(-\frac{1}{2}, 1\right)$:

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 1 &= -\tfrac{1}{2}\left(x + \tfrac{1}{2}\right) \\ y &= -\tfrac{1}{2}x + \tfrac{3}{4} \end{aligned}$$

(3 marks)

(b) Another point Z has coordinates $(-5, -8)$. Show that points A, B and Z are collinear.

$A = (-3, -4)$, $B = (2, 6)$, $Z = (-5, -8)$.

Equation of AB:

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 6 &= 2(x - 2) \\ y &= 2x - 4 + 6 \\ y &= 2x + 2 \end{aligned}$$

Substitute $Z(-5, -8)$:

$$\begin{aligned}
-8 &= 2(-5) + 2 \\
-8 &= -10 + 2 \\
-8 &= -8 \quad \text{Q.E.D.}
\end{aligned}$$

Therefore A, B and Z are collinear.

Total 9 marks

5. (a) The frequency table below shows the number of goals scored in a series of football matches.

Number of Goals scored	0	1	2	3	4	5
Number of Matches	3	10	k	15	4	3

If the mean number of goals is 2.2, determine k.

$$\begin{aligned}
\text{Mean} &= \frac{\Sigma fx}{\Sigma f} \\
2.2 &= \frac{0(3)+1(10)+2k+3(15)+4(4)+5(3)}{3+10+k+15+4+3} \\
2.2 &= \frac{86+2k}{35+k} \\
2.2(35+k) &= 86+2k \\
77+2.2k &= 86+2k \\
0.2k &= 9 \\
k &= 45
\end{aligned}$$

(3 marks)

(b) Teacher Joy recorded the time taken for the 60 students in her Math Club to complete a 12-piece puzzle. The cumulative frequency graph shows information on the time taken.

[Cumulative frequency curve — see original paper]

(i) Probability that a student chosen at random takes at least 42 seconds.

From the curve, cumulative frequency at 42 seconds is 15.

$$\text{Number taking at least 42 s} = 60 - 15 = 45$$
$$P = \frac{45}{60} = 0.75 \text{ or } 75\%$$

(2 marks)

(ii) Use the curve to determine the semi-interquartile range.

$Q_1 = \dfrac{n+1}{4} = \dfrac{60+1}{4}$

Q_1 is the 15.25th value ≈ 42 s (from graph)

$Q_3 = \dfrac{3}{4}(n+1) = \dfrac{3}{4}(61)$

Q_3 is the 45.75th value ≈ 64 s (from graph)

$$\text{Semi-interquartile range} = \frac{Q_3 - Q_1}{2}$$
$$= \frac{64 - 42}{2} = 11 \text{ s}$$

(2 marks)

(iii) Complete the table below.

Time Taken (s)	Number of Students (f)	Midpoint (x)	fx
21–40	13	30.5	396.5
41–60	28	50.5	1414
61–80	16	70.5	1128
81–100	3	90.5	271.5

2nd interval: $41 - 13 = 28$. Total $= 57$, therefore $57 - (28 + 13) = 16$. Then $28 \times 50.5 = 1414$ and $16 \times 70.5 = 1128$. *(2 marks)*

Total 9 marks

6. (a) A recipe for 20 bite-size cookies uses 175 g flour, 125 g butter and 75 g sugar.

(i) Ratio of flour to butter to sugar, in simplest form.

$$175 : 125 : 75 \quad = \quad 7 : 5 : 3$$

(ii) Amount of flour, butter and sugar needed for 52 cookies.

$$\begin{aligned} \text{Scale factor} &= \frac{52}{20} = 2.6 \\ \text{Flour} &= 175 \times 2.6 = 455 \text{ g} \\ \text{Butter} &= 125 \times 2.6 = 325 \text{ g} \\ \text{Sugar} &= 75 \times 2.6 = 195 \text{ g} \end{aligned}$$

(3 marks)

(b) Cookies are made and sold in packs of 12. A bag of flour contains 12.65 kg.

One bag of flour makes a MAXIMUM of ____ packs of cookies.
The amount of flour left over is ____ g.

$$\begin{aligned} \text{Flour for 12 cookies} &= \frac{12}{20} \times 175 = 105 \text{ g} \\ 12.65 \text{ kg} &= 12650 \text{ g} \\ \text{Max number of packs} &= 12650 \div 105 \\ &= 120 \text{ remainder } 50 \end{aligned}$$

One bag of flour makes a MAXIMUM of **120** packs of cookies.
The amount of flour left over is **50** g. *(4 marks)*

Total 9 marks

7. (a) The diagram below shows part of a number chart. The numbers on the chart follow various patterns. Study the patterns and answer the questions that follow.

		Column		
		1	2	3
Row	1	7	9	11
	2	14	16	18
	3	21	23	25
	4	28	30	32
	5	35	37	39
	6	42	44	46

(i) What number should be placed in Column 1, Row 9?

In Column 1, the numbers are multiples of 7.

$$\text{Row 9, Column 1} = 7 \times 9 = 63$$

(1 mark)

(ii) The following diagram represents a given row from the chart.

	Column		
	1	2	3
Row 93	651	653	655

Fill in the three missing numbers in the diagram above.

$$\begin{aligned} \text{Column 1:} &\quad 653 - 2 = 651 \\ \text{Row Number:} &\quad 651 \div 7 = 93 \\ \text{Column 3:} &\quad 653 + 2 = 655 \end{aligned}$$

(2 marks)

(iii) Which row would contain the number 942? Explain how you arrived at your answer.

Row number: 134.

Explanation: Every entry in the chart can be written as $7r + c$ where r is the row number and c is 0, 2 or 4 (the column offset). $942 \div 7 = 134$ remainder 4, so 942 sits in row 134, column 3.

Check:

$$\begin{aligned} \text{Column 1 value} &= 134 \times 7 = 938 \\ \text{Column 2 value} &= 938 + 2 = 940 \\ \text{Column 3 value} &= 940 + 2 = 942 \end{aligned}$$

(2 marks)

(b) A sequence is formed using diagrams made up of equilateral triangles, each constructed from sticks of unit length.

Diagram	Triangles (E)	Sticks (S)	Dots (D)
1	1	3	3
2	3	7	5
3	5	11	7
4	7	15	9
...	...	...	...
n	$2n-1$	$4n-1$	$2n+1$

(i) Complete the row corresponding to diagram n (see table above).

(3 marks)

(ii) Show that the difference between the number of equilateral triangles and the number of dots will always be equal to 2.

$$\begin{aligned} D - E &= (2n+1) - (2n-1) \\ &= 2n + 1 - 2n + 1 \\ &= 2 \end{aligned}$$

Therefore, the difference is always 2. *(2 marks)*

Total 10 marks

SECTION II

Answer ALL questions.

ALGEBRA, RELATIONS, FUNCTIONS AND GRAPHS

8. A quadratic function is defined by $f(x) = 5 + 4x - 2x^2$.

(a) Write the function in the form $f(x) = a(x + h)^2 + k$.

$$\begin{aligned} f(x) &= 5 + 4x - 2x^2 \\ &= -2(x^2 - 2x) + 5 \\ &= -2(x^2 - 2x + 1) + 5 + 2 \\ &= -2(x - 1)^2 + 7 \end{aligned}$$

In the form $a(x + h)^2 + k$ where $a = -2$, $h = -1$ and $k = 7$. *(2 marks)*

(b) A table of values for $f(x) = 5 + 4x - 2x^2$ is shown below.

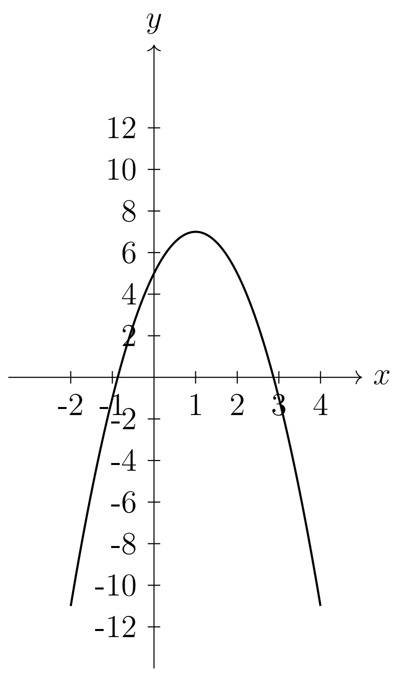
x	-2	-1	0	1	2	3	4
y	-11	-1	5	7	5	-1	-11

(i) Insert the missing values in the table above.

$$\begin{aligned} &\text{When } x = 1: \quad f(x) = 5 + 4(1) - 2(1)^2 = 9 - 2 = 7 \\ &\text{When } x = 3: \quad f(x) = 5 + 4(3) - 2(3)^2 = 17 - 18 = -1 \end{aligned}$$

(2 marks)

(ii) Plot the graph of $f(x) = 5 + 4x - 2x^2$ for $-2 \le x \le 4$, using a scale of 2 cm to 1 unit on the x-axis and 1 cm to 2 units on the y-axis.



[Parabola of $f(x) = 5 + 4x - 2x^2$ — see original paper for grid scale]

(5 marks)

(c) (i) Draw an appropriate line on the graph and use it to estimate the solutions of $5 + 4x - 2x^2 = 3$.

Solutions: $x = -0.4$ and $x = 2.4$ (from graph). *(3 marks)*

Total 12 marks

GEOMETRY AND TRIGONOMETRY

9. The diagram below shows quadrilaterals H, P and N. Quadrilaterals P and N are images of Quadrilateral H after two different transformations.

[Quadrilateral grid with H, P, N — see original paper]

(a) Describe fully the single transformation that maps

(i) Quadrilateral H onto Quadrilateral N.

Quadrilateral H is mapped onto Quadrilateral N by a rotation of 90° anticlockwise about the origin. *(3 marks)*

(ii) Quadrilateral H onto Quadrilateral P.

Quadrilateral H is mapped onto Quadrilateral P by a translation using the vector $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$.

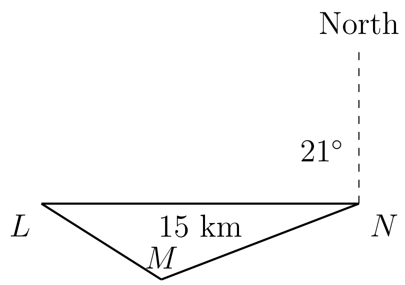
(2 marks)

(b) On the grid, draw the image of Quadrilateral H after it undergoes the following transformations.

(i) An enlargement with scale factor 2 about the centre $(0, -1)$. *Label this image* M. (See grid.) *(2 marks)*

(ii) A reflection in the line $y = -x$. *Label this image* L. (See grid.) *(2 marks)*

(c) The diagram shows three pillars, L, M and N, on level ground. $MN = 12$ **km and** $LN = 15$ **km.** L **is due west of** N**. The bearing of** M **from** N **is** $291°$**.**



Calculate:

(i) Angle MNL.

$$\text{Bearing of } M \text{ from } N = 291^\circ$$
$$\text{Bearing of } L \text{ from } N = 270^\circ$$
$$\angle MNL = 291^\circ - 270^\circ = 21^\circ$$

(1 mark)

(ii) The length of LM.

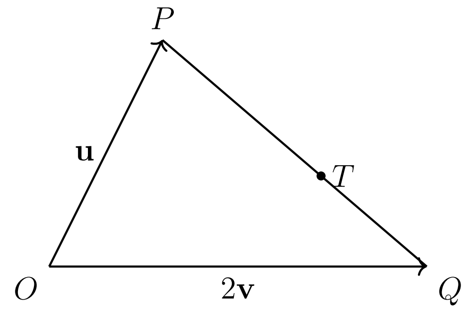
$$
\begin{aligned}
a^2 &= b^2 + c^2 - 2bc\cos A \\
a^2 &= (15)^2 + (12)^2 - 2(15)(12)\cos(21^\circ) \\
a^2 &= 32.91 \\
a &= 5.74 \text{ km}
\end{aligned}
$$

(2 marks)

Total 12 marks

VECTORS AND MATRICES

10. (a) The diagram below shows a scalene triangle, OPQ, with vectors $\vec{OP} = \mathbf{u}$ and $\vec{OQ} = 2\mathbf{v}$. The point T is positioned on the line PQ such that $PT : TQ = 3 : 2$.



Derive a simplified expression, in terms of $\mathbf{u}$ and $\mathbf{v}$, for

(i) $\vec{PQ}$**.**

$$\begin{aligned}\vec{PQ} &= \vec{PO} + \vec{OQ}\\ &= -\mathbf{u} + 2\mathbf{v}\\ &= 2\mathbf{v} - \mathbf{u}\end{aligned}$$

(1 mark)

(ii) $\vec{OT}$**.**

$$\begin{aligned}\vec{OT} &= \vec{OP} + \tfrac{3}{5}\vec{PQ}\\ &= \mathbf{u} + \tfrac{3}{5}(2\mathbf{v} - \mathbf{u})\\ &= \mathbf{u} + \tfrac{6}{5}\mathbf{v} - \tfrac{3}{5}\mathbf{u}\\ &= \tfrac{2}{5}\mathbf{u} + \tfrac{6}{5}\mathbf{v}\end{aligned}$$

(2 marks)

(b) (i) Write down the 2×2 matrices that represent the following transformations.

a) An enlargement with centre $(0, 0)$ and a scale factor of 8.

$$\begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix}$$

b) A reflection in the x-axis.

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

c) An enlargement with centre $(0,0)$ and scale factor 8, followed by a reflection in the x-axis.

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 0 & -8 \end{pmatrix}$$

(4 marks)

(ii) The matrices R and Q are expressed in terms of the constants k and c.

$$R = \begin{pmatrix} 6 & 1 \\ 4 & 2 \end{pmatrix}, \quad Q = \begin{pmatrix} k & 1 \\ c & -6 \end{pmatrix}$$

a) Calculate the matrix product RQ in terms of k and c.

$$\begin{aligned} RQ &= \begin{pmatrix} 6 & 1 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} k & 1 \\ c & -6 \end{pmatrix} \\ &= \begin{pmatrix} 6k + c & 6 - 6 \\ 4k + 2c & 4 - 12 \end{pmatrix} \\ &= \begin{pmatrix} 6k + c & 0 \\ 4k + 2c & -8 \end{pmatrix} \end{aligned}$$

(2 marks)

(iii) RQ represents the transformation of an enlargement with centre $(0,0)$ and a scale factor of 8, followed by a reflection in the x-axis. Determine the values of k and c.

$$\begin{pmatrix} 6k + c & 0 \\ 4k + 2c & -8 \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 0 & -8 \end{pmatrix}$$

Thus:

$$\begin{aligned} 6k + c &= 8 \quad \text{Eq 1} \\ 4k + 2c &= 0 \quad \text{Eq 2} \end{aligned}$$

Eq 1 $\times 2$: $12k + 2c = 16$ (Eq 3)

Eq 3 $-$ Eq 2:

$$\begin{aligned} 8k &= 16 \\ k &= 2 \end{aligned}$$

Substitute into Eq 1:

$$\begin{aligned} 6(2) + c &= 8 \\ c &= 8 - 12 \\ c &= -4 \end{aligned}$$

(3 marks)

Total 12 marks

END OF TEST